Structural Aliasing Analysis in Patch-Based Time-Series Foundation Models

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May 13, 2026

Executive Summary

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Introduction

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$$\cos^3\theta = \frac{1}{4}\cos\theta + \frac{3}{4}\cos 3\theta \qquad (1)$$

During the preparation of this work, the authors used AI-based tools, including GPT-4o, GPT-5.3, and GPT-5.4, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Automatically referencing an equation number using its label: Equation 1.

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Title 1

Setup and Formal Definition

Let $x[n]$ be a discrete-time signal with $n \in \mathbb{Z}$, sampled at frequency f_s. Patch-based tokenisation extracts a sequence of tokens by sliding a window of size P samples with stride S:

$$\mathbf{p}_k = \big(x[kS],\ x[kS+1],\ \ldots,\ x[kS+P-1]\big) \in \mathbb{R}^P, \qquad k = 0, 1, 2, \ldots \tag{2}$$

Definition 1 (Patch Tokenisation Operator)**.** *The map $\mathcal{T} : \ell^\infty(\mathbb{Z}) \to (\mathbb{R}^P)^{\mathbb{N}}$ defined by (1) is called the* patch tokenisation operator. *It is linear: collecting K patches yields*

$$\mathbf{P} = \mathcal{T}\{x\} = \mathbf{W} \cdot \mathbf{x}, \tag{3}$$

where $\mathbf{W} \in \{0,1\}^{KP \times N}$ is a binary selection matrix whose k-th block of P rows picks the P consecutive samples starting at index kS.

Condition for Two Consecutive Patches to Be Identical

Proposition 1. *Patches $\mathbf{p}_k$ and $\mathbf{p}_{k+1}$ are identical if and only if*

$$x[n] = x[n+S] \quad \forall\, n \in \{kS, kS+1, \ldots, kS+P-1\}. \tag{4}$$

Globally, if the signal is periodic with period T_0 (in samples), identical consecutive patches occur whenever

$$\boxed{S = m \cdot T_0, \qquad m \in \mathbb{Z}^+.} \tag{5}$$

Equivalently, in terms of signal frequency f and sampling frequency f_s:

$$f = m \cdot \frac{f_s}{S}, \qquad m \in \mathbb{Z}^+. \tag{6}$$

Proof. $\mathbf{p}_k = \mathbf{p}_{k+1}$ component-wise requires $x[kS+j] = x[(k+1)S+j]$ for all $j \in \{0, \ldots, P-1\}$, i.e. $x[n] = x[n+S]$ on that window. For a periodic signal with $T_0 = f_s/f$ samples, this holds whenever S is an integer multiple of T_0, giving (4). □

Loss of Observability

When $\mathbf{p}_k = \mathbf{p}_{k+1}$ for all k, the token sequence is constant — the model sees the same vector repeated. The *rank* of the token matrix $\mathbf{P}$ collapses to 1. Consequently:

- Temporal variation is invisible to the model.
- Any two signals differing only by a phase shift of S samples produce the same token sequence: they are **observationally equivalent**.
- Even when patches are not fully identical but merely highly correlated ($S \approx mT_0$), the effective information in the token sequence is reduced (*soft* loss of observability).

Title 2

Setup and Formal Definition

Let $x[n]$ be a discrete-time signal with $n \in \mathbb{Z}$, sampled at frequency f_s. Patch-based tokenisation extracts a sequence of tokens by sliding a window of size P samples with stride S:

$$\mathbf{p}_k = \big(x[kS],\ x[kS+1],\ \ldots,\ x[kS+P-1]\big) \in \mathbb{R}^P, \qquad k = 0, 1, 2, \ldots \tag{7}$$

Definition 2 (Patch Tokenisation Operator)**.** *The map $\mathcal{T} : \ell^\infty(\mathbb{Z}) \to (\mathbb{R}^P)^{\mathbb{N}}$ defined by (1) is called the* patch tokenisation operator*. It is linear: collecting K patches yields*

$$\mathbf{P} = \mathcal{T}\{x\} = \mathbf{W} \cdot \mathbf{x}, \tag{8}$$

where $\mathbf{W} \in \{0,1\}^{KP \times N}$ is a binary selection matrix whose k-th block of P rows picks the P consecutive samples starting at index kS.

Condition for Two Consecutive Patches to Be Identical

Proposition 2. *Patches $\mathbf{p}_k$ and $\mathbf{p}_{k+1}$ are identical if and only if*

$$x[n] = x[n+S] \quad \forall\, n \in \{kS,\, kS+1,\, \ldots,\, kS+P-1\}. \tag{9}$$

Globally, if the signal is periodic with period T_0 (in samples), identical consecutive patches occur whenever

$$\boxed{S = m \cdot T_0, \qquad m \in \mathbb{Z}^+.} \tag{10}$$

Equivalently, in terms of signal frequency f and sampling frequency f_s:

$$f = m \cdot \frac{f_s}{S}, \qquad m \in \mathbb{Z}^+. \tag{11}$$

Proof. $\mathbf{p}_k = \mathbf{p}_{k+1}$ component-wise requires $x[kS+j] = x[(k+1)S+j]$ for all $j \in \{0, \ldots, P-1\}$, i.e. $x[n] = x[n+S]$ on that window. For a periodic signal with $T_0 = f_s/f$ samples, this holds whenever S is an integer multiple of T_0, giving (4). □

Loss of Observability

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- Even when patches are not fully identical but merely highly correlated ($S \approx mT_0$), the effective information in the token sequence is reduced (*soft* loss of observability).

Example

Methodologies

Sample Sites & Processing

This line shows how to use a footnote to further explain or cite text[1].

This is a bullet point list:

- Arcu eros accumsan lorem, at posuere mi diam sit amet tortor
- Fusce fermentum, mi sit amet euismod rutrum
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Species Identification

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Data Analysis

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[1] Example footnote text.

Table 1: *Example single column table.*

Location		
East Distance	West Distance	Count
100km	200km	422
350km	1000km	1833
600km	1200km	890

Table 2: *Example two column table with fixed-width columns.*

Location		
East Distance	West Distance	Count
100km	200km	422
350km	1000km	1833
600km	1200km	890

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Results

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Figure 1: *Anther of thale cress (Arabidopsis thaliana), fluorescence micrograph. Source: Heiti Paves, https://commons.wikimedia.org/wiki/File:Tolmukapea.jpg.*

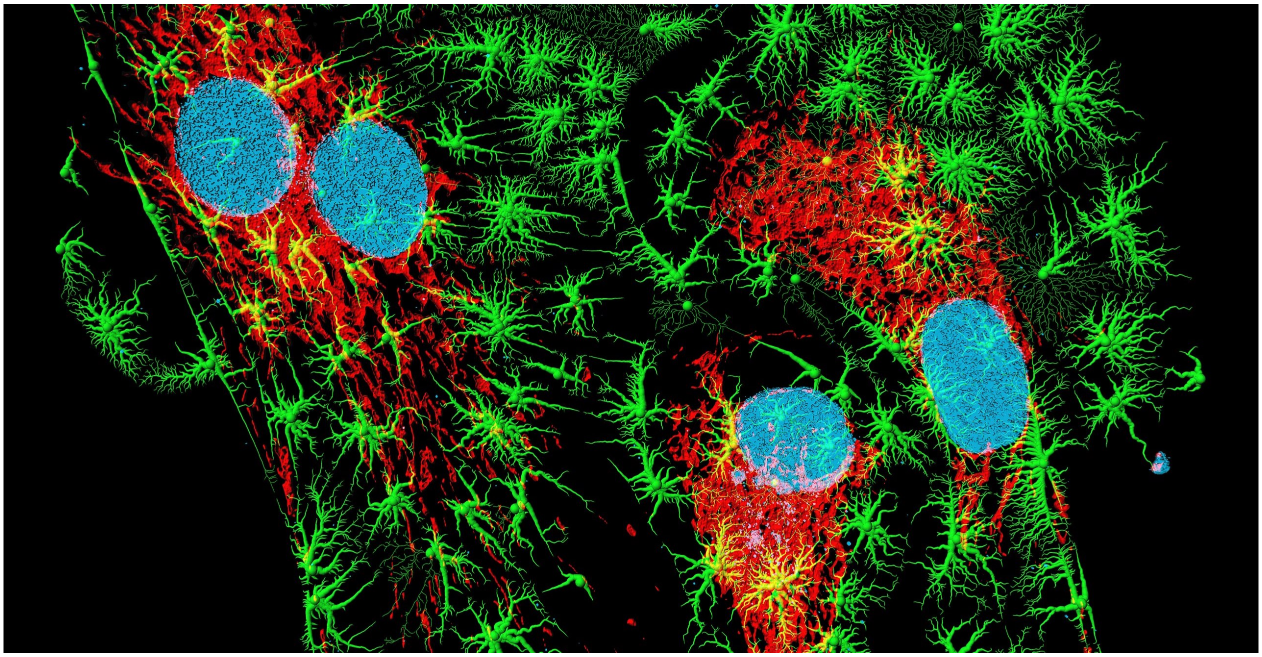

Figure 2: *Bovine pulmonary artery endothelial cells in culture. Blue: nuclei; red: mitochondria; green: microfilaments. Computer generated image from a 3D model based on a confocal laser scanning microscopy using fluorescent marker dyes. Source: Heiti Paves, https://commons.wikimedia.org/wiki/File:Fibroblastid.jpg.*

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International Support

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Links

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Discussion

This statement requires citation [1]. This statement requires multiple citations [2]. This statement contains an in-text citation, for directly referring to a citation like so: Voita and Titov [3].

Subsection One

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Subsection Two

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References

[1] *Fast and Accurate Zero-Shot Forecasting with Chronos-Bolt and AutoGluon | Artificial Intelligence*. Dec. 2, 2024. URL: `https://aws.amazon.com/blogs/machine-learning/fast-and-accurate-zero-shot-forecasting-with-chronos-bolt-and-autogluon/` (visited on 05/13/2026).

[2] Nora Belrose et al. "LEACE: Perfect Linear Concept Erasure in Closed Form". In: *Advances in Neural Information Processing Systems* 36 (Dec. 15, 2023), pp. 66044–66063. URL: `https://proceedings.neurips.cc/paper_files/paper/2023/hash/d066d21c619d0a78c5b557fa3291a8f4-Abstract-Conference.html` (visited on 05/13/2026).

[3] Elena Voita and Ivan Titov. "Information-Theoretic Probing with Minimum Description Length". In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. EMNLP 2020. Ed. by Bonnie Webber et al. Online: Association for Computational Linguistics, Nov. 2020, pp. 183–196. DOI: `10.18653/v1/2020.emnlp-main.14`. URL: `https://aclanthology.org/2020.emnlp-main.14/` (visited on 05/13/2026).